

ELECTRICITY

Electricity is used as a way of moving energy from place to place. It can take energy from burning a fuel, such as coal or gas, in a power station into your home to work your television. Most electricity is generated in machines. Small machines, called dynamos, light the lights on bikes.



Dynamo

Huge generators in power stations light whole cities. Pedal power works a dynamo, but steam produces the electricity in a power station.

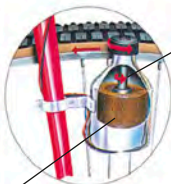
This steam is made by using the heat from burning fossil fuels or splitting atoms.

Sunshine, falling water, and whirling windmills can also generate electricity.



Mighty machine

The big blue generator inside this power station is about ten times as tall as you!



This magnet spins around because it is fixed to a rod that touches the turning wheel.

Electricity is generated in this coil of wire by the spinning magnet.

A pile of coal as heavy as 40 elephants is burned each hour.

Choice of fuels

A power station uses just one sort of fuel to generate its electricity.

This one burns coal, but other power stations use oil, natural gas, or nuclear fuel.

Oil



Natural gas



Uranium

1. Coal is crushed and then blown into the boiler to burn.

2. The burning coal makes water turn into steam.

3. Steam surges from the boiler into the generator. It pushes around huge paddles, called turbines.

Chimney

4. Turbines turn a massive magnet 50 times per second.

5. The moving magnet creates an electric current in huge coils of wire.

First transformer

A condenser turns the hot steam into hot water.

The cooling tower cools the hot water so that it can be used again.

